

Developing and validating clinical prediction models in hepatology – An overview for clinicians

Rickard Strandberg^{1,*}, Peter Jepsen^{2,3}, Hannes Hagström^{1,4}

Summary

Prediction models are everywhere in clinical medicine. We use them to assign a diagnosis or a prognosis, and there have been continuous efforts to develop better prediction models. It is important to understand the fundamentals of prediction modelling, thus, we herein describe nine steps to develop and validate a clinical prediction model with the intention of implementing it in clinical practice: Determine if there is a need for a new prediction model; define the purpose and intended use of the model; assess the quality and quantity of the data you wish to develop the model on; develop the model using sound statistical methods; generate risk predictions on the probability scale (0-100%); evaluate the performance of the model in terms of discrimination, calibration, and clinical utility; validate the model using bootstrapping to correct for the apparent optimism in performance; validate the model on external datasets to assess the generalisability and transportability of the model; and finally publish the model so that it can be implemented or validated by others.

© 2024 The Author(s). Published by Elsevier B.V. on behalf of European Association for the Study of the Liver. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).



Introduction

Clinical prediction models are common across all fields of medicine. They are tools to support clinical decisions about a patient's diagnosis or prognosis, and the model inputs are usually patient characteristics, e.g. a blood test result or the patient's age. The model output is sometimes a probability of having a specific diagnosis or prognosis, and sometimes it is just a number (a 'score') which may not translate into a probability, but nevertheless assists with clinical decision making.

The Simplified Diagnostic Criteria for Autoimmune Hepatitis is an example of a diagnostic prediction model.¹ It combines information from blood tests and liver biopsy into a score that classifies the patient as having 'probable autoimmune hepatitis', 'definitive autoimmune hepatitis', or neither. The model for end-stage liver disease (MELD) score is an example of a prognostic prediction model. It was developed to predict 90-day mortality in patients with cirrhosis following insertion of a transjugular intrahepatic portosystemic shunt (TIPS). Originally, the model output was predicted risk of death within 90 days,² but the inputs and outputs from the MELD score have since evolved, so that the output of today's MELD score is not predicted risk of 90-day mortality, but a score with implicit information about the patient's risk of death. This information is then used to make decisions on the need and timing of an intervention, such as listing for liver transplantation. Hence, prediction models help clinicians make better decisions, but as

we will see, prediction models may be harmful if they are not developed properly.

When we discuss prediction models, we refer to both diagnostic and prognostic models. The core concept is to use various measurements and other information available about the patient (so-called predictors) and estimate a risk (of having a specific diagnosis or prognosis) for that specific patient at that specific time. Depending on the predicted risk, the patient and healthcare provider can make shared decisions such as initiating an intervention or treatment, or deciding on the need for additional testing. Note that sometimes there might be nuances to the prediction problem which make it difficult to immediately classify the model as either diagnostic or predictive, for example when one wants to assign treatment based on the predicted response to that treatment.

There is an inherent value in studying and comparing multiple predictors and estimating their relative effects. Indeed, the statistical approach of prediction modelling is practically indistinguishable from a multivariable analysis, or from studying a specific association while controlling for multiple confounders. The difference is that in a prediction model we do not have a special interest in any one particular predictor, nor do we care whether the association with the outcome is causal or just a correlation.³ Prediction models are concerned with correctly classifying who will and will not experience the outcome or event of interest, and with producing accurate risk estimates for the individual patient. These risk estimates are

Keywords: Diagnosis; Prognosis; Hepatology; Modeling; Prediction; Discrimination; Calibration; Validation; Utility.

Received 30 October 2023; received in revised form 6 March 2024; accepted 12 March 2024; available online 24 March 2024

* Corresponding author. Address: C1:77, Division of Hepatology, Karolinska University Hospital, 141 86 Stockholm, Sweden. (R. Strandberg)

E-mail address: rickard.strandberg@ki.se (R. Strandberg).

<https://doi.org/10.1016/j.jhep.2024.03.030>



ELSEVIER

Key points:

We describe nine steps to develop and validate a clinical prediction model with the intention of implementing it in clinical practice:

1. Determine if there is a need for a new prediction model.
2. Define the purpose and intended use for the model.
3. Assess the quality and quantity of the data you wish to develop the model on.
4. Develop the model using sound statistical methods.
5. Generate risk predictions on the probability scale (0-100%).
6. Evaluate the performance of the model in terms of discrimination, calibration, and clinical utility.
7. Validate the model using bootstrapping to correct for the apparent optimism in performance.
8. Validate the model on external datasets to assess the generalisability and transportability of the model.
9. Publish the model so that it can be implemented or validated by others.

usually presented as probabilities between 0% and 100% that the event of interest has occurred (or will occur within a given time). Based on the predicted probability, a decision can then be made on which intervention patients should receive, such as treatment or further testing.

Making a diagnosis or prognosis for patients is a key part of practicing medicine. It is therefore understandable that the development of prediction models is a popular research area. Unfortunately, there are several pitfalls in the development of prediction models, and with the current inflation in published prediction models it can be difficult to separate the wheat from the chaff.

We will emphasise the importance of having a pre-defined clinical purpose when considering developing a prediction model, and for there to be a viable path for actual implementation in practice. In diagnosing chronic liver diseases, non-invasive approaches to replace liver biopsy are attractive.⁴ An example of a non-invasive test is the Fibrosis 4 (FIB-4) score, aimed at diagnosing the presence of advanced liver fibrosis.⁵ This purpose falls under the umbrella of prediction, since we are using biomarkers and other variables to predict a current disease state — in this case advanced fibrosis.

Our aim with this review is to introduce the clinical reader to the development and validation of prediction models. We outline nine steps in the development of a new clinical prediction model. The emphasis will be on careful planning, sound model development methods, rigorous validation of performance, and transparent reporting. We will assume that we are working with a diagnostic model with a binary outcome (does the patient have the disease: yes or no). This is the most common type of outcome for prediction in medicine. At the end we will include some additional considerations for prognostic models with a time-to-event outcome. We will give real examples of both diagnostic and prognostic models.

Example case study: predicting the presence of cirrhosis in primary biliary cholangitis (PBC)

To illustrate the steps involved in developing and validating a clinical prediction model, we will use the Mayo Clinic PBC dataset,⁶ which is old but publicly available as part of the R statistical software. The data and all code used in this paper to

develop a prediction model are publicly available at <https://github.com/rickstra/PBCexample>.

A total of 312 patients with PBC between the years 1974 and 1984 were part of a randomised-controlled trial of D-penicillamine. For our purposes, the treatment will be ignored (it was found to be ineffective), and the two arms will be combined into one. Our modelling example will be a diagnostic model, so the assigned treatment at baseline is irrelevant.

The variables recorded in the dataset are described in Table 1. For simplicity, we have imputed the missing values using a single imputation with predictive mean matching, and then treated the dataset as if complete.⁷ However, in practice, multiple imputation should be used instead.⁷

1) Search the literature and consider the need for a new prediction model

Before embarking on the development of a new clinical prediction model, we must make sure that it is needed. If a model already exists, especially if it is already in clinical use or recommended by guidelines, we should consider validating that model on or recalibrating it to our data instead.⁸ The MELD score is an example of a model which has been both repurposed and updated. It was originally developed to predict 3-month survival in TIPS recipients.² It was subsequently modified (disease aetiology removed) to predict survival in all patients with end-stage cirrhosis.⁹ Over time, the model has been updated twice: once to include serum sodium level,¹⁰ and again to include sex and serum albumin.¹¹ Each time it was trained on a larger dataset.

2) Formulate the problem and define the intended use

We must make it clear what we are trying to predict, and why. The four basic components we need to explicitly define are:

1. An outcome to predict. For example, the presence of a disease, a specific stage or severity of a disease, or the future occurrence of a disease within a certain timeframe (e.g. 5 years).
2. A target population for the prediction model. The data sample we develop our model on should be from a population representative of the target population. For instance, a model developed in patients with advanced disease seen in tertiary

Table 1. Variables in the primary biliary cholangitis data.

	Fibrosis stage 1-3 (n = 200)	Fibrosis stage 4 (n = 88)	Overall (N = 288)
Age (years)			
Median [IQR]	48.6 [40.7–55.5]	52.5 [43.8–58.9]	49.0 [41.4–56.4]
Sex			
Male	21 (10.5%)	12 (13.6%)	33 (11.5%)
Female	179 (89.5%)	76 (86.4%)	255 (88.5%)
Oedema status			
No oedema	182 (91.0%)	75 (85.2%)	257 (89.2%)
Untreated or successfully treated	15 (7.5%)	10 (11.4%)	25 (8.7%)
Oedema despite diuretic therapy	3 (1.5%)	3 (3.4%)	6 (2.1%)
Presence of spiders			
No	161 (80.5%)	50 (56.8%)	211 (73.3%)
Yes	39 (19.5%)	38 (43.2%)	77 (26.7%)
Presence of hepatomegaly			
No	130 (65.0%)	17 (19.3%)	147 (51.0%)
Yes	70 (35.0%)	71 (80.7%)	141 (49.0%)
Platelet count (10 ⁹ /L)			
Median [IQR]	271 [216–336]	237 [165–298]	265 [206–326]
Missing	4 (2.0%)	0 (0%)	4 (1.4%)
Standardised blood clotting time			
Median [IQR]	10.3 [9.90–10.8]	11.0 [10.2–11.6]	10.5 [10.0–11.0]
Serum albumin (g/dl)			
Median [IQR]	3.65 [3.43–3.88]	3.44 [3.19–3.64]	3.57 [3.35–3.82]
Alkaline phosphatase (U/L)			
Median [IQR]	1,190 [856–1,870]	1,390 [955–2,140]	1,250 [858–1,970]
Aspartate aminotransferase (U/ml)			
Median [IQR]	109 [76.0–142]	121 [89.9–156]	113 [79.7–150]
Serum bilirubin (mg/dl)			
Median [IQR]	1.10 [0.70–2.33]	2.25 [1.10–5.03]	1.30 [0.70–3.20]
Serum cholesterol (mg/dl)			
Median [IQR]	316 [256–420]	304 [251–380]	315 [253–406]
Missing	12 (6.0%)	13 (14.8%)	25 (8.7%)
Urine copper (µg/day)			
Median [IQR]	63.0 [37.0–99.0]	92.5 [57.8–139]	69.0 [40.5–115]
Missing	1 (0.5%)	0 (0%)	1 (0.3%)
Triglycerides (mg/dl)			
Median [IQR]	109 [83.5–152]	101 [84.5–136]	107 [84.0–146]
Missing	13 (6.5%)	13 (14.8%)	26 (9.0%)

care hospitals may not work in a primary care population (and vice versa).

3. An intervention which will be applied based on the prediction. This could be a direct treatment of some kind, or it could be further testing or monitoring (if the prediction model is to be used as a screening tool).
4. A potential beneficial impact of implementing the prediction model. For example, it may enable identification of patients in need of treatment (thus improving their health outcomes) or reduce the number of unnecessary treatments.

In our illustrative example, we will aim to develop a diagnostic model to identify patients with PBC and fibrosis stage 4 (cirrhosis). We wish to specifically identify such patients without the need for a (hypothetical) liver biopsy. The target population for this prediction model is all patients with PBC. However, if a patient presents with ascites, then testing with the model is redundant since it is then apparent that the patient is likely to have cirrhosis, so we exclude those patients from the target population. We imagine that the treatment/intervention will be a confirmatory biopsy. If the biopsy shows cirrhosis, the patient can be put under surveillance for hepatocellular carcinoma and screened for oesophageal varices, if appropriate. Thus, if our

prediction model works well (Steps 6–8), we could reduce the number of biopsies needed among patients with PBC.

Another example is the FIB-4 score. It was originally developed to predict advanced liver fibrosis (fibrosis stage 3–4) in patients with both HIV and hepatitis C. It has since been applied to several other settings, e.g. it has been recommended as a tool to rule out advanced fibrosis in patients with metabolic dysfunction-associated steatotic liver disease (MASLD).¹² A better prediction model might be developed specifically for the new application via “re-calibration” – even with the exact same predictors.

3) Inspect the data and determine the scope

Once we have defined the problem that our prediction model aims to solve, we can turn to the data. We want to inspect the data for both quality and quantity. This will give us an idea of how we should develop our model and how reliable it will be.

First, we want the dataset to include all the predictors which are of interest in clinical practice.¹³ We would not want to present a prediction model without a particular predictor simply because we did not have that predictor in our dataset. For example, a model to predict patients’ future risk of

hepatocellular carcinoma that is developed using a dataset without information on presence of cirrhosis is very unlikely to be adopted by clinicians.

We should consider the source of each potential predictor. Some predictors are effectively “free”, such as age, sex, weight, and family history of disease. Other predictors require a physical examination by a clinician, or a blood sample. Some predictors require more advanced measuring methods, e.g. polygenic risk scores, “omics”-data, or imaging, such as liver stiffness measures based on transient elastography. Such predictors can be too costly depending on the scale of the intended use case. For instance, a model intended for use in primary care may never be used if it contains predictors that are too costly to measure. By considering the source of each predictor, we get a head start in choosing which of the predictors offered to us by the data will be feasible to use in practice. This increases the probability that the model will see clinical use.

In our PBC example, the “free” predictors are age and sex. Predictors such as platelets, prothrombin time, albumin, bilirubin, alkaline phosphatase, aspartate aminotransferase, cholesterol, and triglycerides require a blood sample, while presence of oedema, spider nevi, and hepatomegaly require a clinical assessment; and finally copper level requires a urine sample. In our intended use case, where we wish to reduce the number of biopsies, we are willing to make use of any predictor we can. We choose to omit urine copper since it is the only predictor requiring a urine sample. We also exclude prothrombin time since it is a test which is no longer in regular use in many settings. Also recall that patients with ascites are excluded from the target population, and as such it will not be a predictor. To make a prediction using our model, we will therefore require a clinical assessment and a blood sample.

When it comes to the amount of data we have available for a prediction model, we consider the “effective sample size”, which is determined by the number of events in a dataset rather than its total size (e.g. 88 events of cirrhosis in the PBC dataset, not the 288 total patients). A sample size of 10,000 patients is more useful when there are 3,000 events than if there are only 10 events, for example. In a diagnostic model with a binary outcome, the effective sample size is either the number of events or the number of non-events, whichever is smallest (e.g. the effective sample size is 3,000 if there are 7,000 events among 10,000 patients). In a prognostic model the effective sample size is determined by the number of events of the outcome of interest.

The effective sample size we need is determined by the complexity of the prediction model. The number of “degrees of freedom” of a model is the total number of parameters we add into our model which need to be estimated. The more degrees of freedom we require, the more data we need to accurately estimate all parameters at once. The term comes from the number of degrees of freedom used when testing the statistical significance of the entire model against a null intercept-only model and should not be confused with the residual degrees of freedom from ordinary linear regression or analysis of variance.

Each predictor requires at least one degree of freedom, but potentially more. For a categorical predictor, the degrees of freedom are the number of categories minus one. For example, the presence of oedema in our PBC example has three

categories (no oedema; oedema responsive to treatment; oedema not responsive to treatment) and thus requires two degrees of freedom. If we want to allow a non-linear relationship between a continuous predictor and the outcome (e.g. using square terms or spline functions) additional degrees of freedom are required depending on the complexity of the desired non-linearity. Interactions between predictors also require additional degrees of freedom.

A common rule of thumb in the past has been to have at least 10 events per degree of freedom required in the model.^{14–16} By this rule, our PBC dataset with an effective sample size of 88 would be allowed nine degrees of freedom, for example. Recently, a more targeted approach has been developed.¹⁷ Depending on the fraction of events and a pre-specified model complexity (number of total degrees of freedom for all candidate predictors), we can estimate the minimum sample size needed to fulfil four criteria for good prediction performance (where sample size $\times$ event fraction $\approx$ effective sample size). The criteria are:

1. **Precisely estimate the overall risk.** As a first criterion we should be able to estimate the event proportion with a certain precision, such as within a margin of error of $\pm 5\%$ points. For example, if the true event proportion is 30%, we want the 95% CI for our estimate of the event proportion to be within 25–35%.
2. **Small mean absolute prediction error.** The absolute prediction error is the absolute difference between a patient’s predicted risk and their true risk. The true risk is never known, but an approximate formula can still be used to estimate the sample size needed to keep the mean absolute prediction error sufficiently small (e.g. at less than 5%).
3. **Small estimated “shrinkage factor.”** The so-called heuristic shrinkage factor proposed by Van Houwelingen and Le Cessie¹⁸ is a way to compensate for a model being too complex for the amount of data it is trained on. Its value is between 0 and 1. To get better predictions in new patients, we can multiply our prediction model’s parameter estimates with this number to make the predictions less extreme (we “shrink” the estimates towards 0). We want enough sample size to limit the amount of recommended shrinkage (shrinkage factor above 0.9).
4. **Small expected optimism of the model fit.** The estimated performance of a prediction model will be better in the data sample it was developed in than it will be in new patients from the same target population. We say that this apparent performance is optimistic. The more data we have, the smaller the optimism will be. We want to have enough of a sample size to limit the expected optimism. See Step 7 for more details.

The theory and calculations behind these criteria are beyond the scope of this paper, but we strongly recommend using them. We have included the formulas for logistic regression models in the [supplementary materials](#), together with a simplified figure for easy reference.

Oftentimes, data collection is carried out before the prediction model is developed. In such cases, the available sample size could be used to instead determine the number of degrees of freedom the model can reasonably contain based on those four criteria with the data that is available.

In [Table 2](#), we see the sample size criteria applied to our PBC example. We have 12 candidate predictors remaining (thus a minimum of 12 degrees of freedom). If we allow the possibility to spend two more degrees of freedom on

Table 2. Sample size criteria applied to our primary biliary cholangitis example.

Criterion	Description	Target value	Sample size	Achieved value	df
1	Precisely estimate the overall risk	<0.05	326	0.053	—
2	Small mean absolute prediction error	<0.05	635	0.077	6.0
3	Small estimated “shrinkage factor”	>0.90	455	0.852	8.9
4	Small expected optimism of the model fit	<0.05	385	0.059	11.9

It shows the target value of each criterion, the sample size needed to achieve it based on using 14 degrees of freedom for the model, the achieved value of the criterion based on our sample size of 288 and 14 degrees of freedom, and the maximum number of degrees of freedom allowed in order to reach the target based on our sample size of 288. d.f, degrees of freedom.

non-linearity or multiple categories, we can decide to make our sample size calculations based on a hypothetical 14 degrees of freedom model. The sample size needed to hit all four targets is 635, which is more than twice our available sample size. We see that it is criterion 2 which demands such a large sample size. With our sample size of 288 we should therefore expect that the mean absolute prediction error will be larger than we would like (0.077 instead of 0.05).

It is common to have missing predictor values for some individuals. The most straight-forward method for handling missing data is to delete any observations with missing predictor values (usually referred to as “complete case” analysis). However, we recommend using a more advanced technique such as multiple imputation^{5,19} to help preserve as much effective sample size as possible, and hopefully avoid the potential bias that deleting incomplete observations can introduce.²⁰

4) Specify and fit the model

When the problem and the data are well understood, we are ready to develop the prediction model. The type of model to use will largely be determined by the outcome we are trying to predict. In a diagnostic model with binary outcome, the logistic regression model is the natural choice. For a prognostic model predicting the 3-year risk, for example, survival models such as cause-specific Cox proportional hazards models,⁶ Fine-Gray subdistribution hazards models,²¹ or flexible parametric survival models^{22,23} are viable choices (see “Additional considerations for prognostic models” below).

Any model fitted to a data sample is overfitted to that sample to some degree. The model will pick up features present in the whole target population, as well as features specific only to the actual sample from that population. This means that a substantially overfitted model will translate poorly outside the particular sample, since it will pick up features only present in the sample. As we shall see (Steps 7-8), the key purpose of validating a prediction model is to expose the amount of overfitting. A simplified illustration of overfitting (and underfitting) can be seen in Fig. 1.

The sample size and degrees of freedom calculations made in Step 3¹⁷ are intended to keep the amount of overfitting in the prediction model in check. This is often easier said than done when we have many predictors and want to allow flexibility in their effects. Further techniques can be applied to combine related or correlated predictors, transform predictors, and eliminate redundant predictors before fitting the prediction model. See¹⁵ for the specifics.

Other advice for producing better prediction models includes not categorising continuous predictors (such as age or platelet count).²⁴ Even if the cut-offs used are to some extent data driven, it still makes strong assumptions about the effect

of the predictor on the outcome. It asserts both that there is no difference in risk for someone at the beginning of the cut-off and someone at the end of the cut-off, and that there is a sharp difference in risk at the very end of one cut-off compared to the very start of the next cut-off – even if the difference in predictor value is very small. For instance, if age is a predictor and we introduce a cut-off at 65 years we would treat individuals aged 66 years at an equal risk as those aged 99 years, which is clearly unreasonable. The categorisation is also sensitive to where the cut-offs are placed, and each added cut-off increases the number of degrees of freedom needed. We must of course acknowledge that categorisation has historically made prediction models easier to use clinically. With better access to computers and calculators, model complexity is less of a hindrance and the convenience of categorisation should not be considered worth sacrificing prediction accuracy for.

If we suspect that the relationship with the outcome is not linear (based on subject-matter knowledge), we should instead allow for non-linear terms in the model, such as through restricted cubic splines¹⁵ or fractional polynomials.²⁵ Both are ways of modelling the relationship between a continuous predictor and the outcome as something more general than just a straight line. That way we can spend multiple degrees of freedom without having the strong assumptions of the fixed cut-offs. Generally, it is better to allow the relationship to be non-linear if nothing else is known.¹⁵

If we have reduced the number of degrees of freedom as much as we can *a priori*, and if we suspect that overfitting will

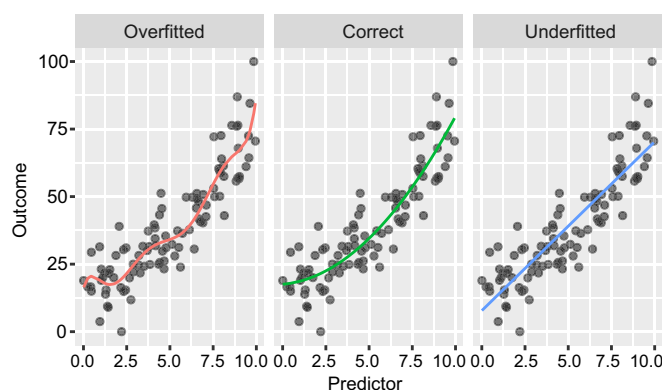


Fig. 1. A simulated illustration of overfitting and underfitting. The points show a slightly non-linear relationship between a predictor and an outcome. The line in each panel corresponds to an attempted model of the relationship. The middle panel shows the correct model (a square polynomial). In the left panel, the model (a 7th degree polynomial) overfits to the data by picking up some of the noise. Conversely, the right panel shows an underfitted model (a straight line) which misses the non-linear feature and particularly underpredicts the outcome for high values of the predictor.

still be a problem, we can turn to shrinkage techniques to reduce the remaining overfitting. We briefly mentioned the heuristic shrinkage factor¹⁸ in the previous step. Other more advanced shrinkage techniques such as penalised maximum likelihood estimation²⁶ or LASSO (least absolute shrinkage and selection operator)^{27,28} exist and are intended to compensate for a model which is too complex for its data. The basic concept behind all shrinkage techniques is to put a constraint (or penalty) on the model coefficients which “shrinks” them towards 0. This introduces a bias in the model’s fit to the training sample which makes the model’s predictions less extreme but reduces the overfitting and on average improves predictions in external datasets. The modelling becomes more complicated with these methods as they require us to specify one (or more) tuning parameters which control the amount of shrinkage to apply. This is typically done using cross-validation or the Akaike information criterion.

When it comes to implementing a prediction model into practice, it might be necessary to include only a few of the predictors under consideration – especially if some predictors are much weaker than others. Variable selection schemes such as univariate screening (fitting each predictor separately and eliminating those with insignificant *p* values) or stepwise selection (alternating between adding and eliminating predictors based on *p* values) should be avoided as they run a high risk of eliminating important predictors and including uninformative predictors. See Chapter 4.3 of¹⁵ for a comprehensive summary of the issues. We have also included a simple simulation in the [supplementary materials](#). Most notably, any variable selection based on *p* values increases overfitting. The estimated coefficients are not the same as if the final model was pre-specified – they are instead the estimated coefficients given that they are statistically significant. This important difference will bias the estimates away from zero, which creates an effect opposite to that of shrinkage (which biases towards zero), *i.e.* it will increase overfitting.

Instead, we should try to use prior knowledge for which variables to include. The assessment we made in Step 3 regarding the requirements for collecting predictors gives us a start. Additionally, we can use our own subject-matter knowledge and previous studies to select or otherwise prioritise certain predictors over others. The same goes for interactions between predictors: if there is room for the interaction based on the complexity-to-data trade-off, and if it is known to be substantive enough, then – and only then – it can be included. If one predictor is known to have a strong association with the outcome, we can assign more degrees of freedom to that predictor (through non-linearity or the number of categories) compared to other predictors with weaker associations (which can be kept linear or reduced to two categories).

A potentially beneficial feature of using LASSO as your shrinkage method is that it will in theory shrink the effects of some (weak) predictors to exactly 0, which effectively becomes a form of variable selection. It performs particularly well when the proportion of non-informative predictors is large, such as when omics data is involved.

In our PBC example, we have 12 potential predictors available to us according to our reasoning in Step 3. We combined the levels of oedema status “Untreated or successfully treated” and “Oedema despite diuretic therapy” (see [Table 1](#)) into one, making it a one degree of freedom predictor. We then allowed the continuous predictors albumin, cholesterol, and bilirubin to

be non-linear, modelled using three-knot restricted cubic splines (which require two degrees of freedom each). This potential model had 15 degrees of freedom total, which is one more than the 14 we initially planned for.

We used penalised regression to shrink the model estimates to compensate for having more degrees of freedom than the sample size calculations recommended. From this penalised model with all predictors, we calculated the predicted risks and fitted an approximated model using ordinary linear regression onto those predicted risks with only five of the predictors which together explained at least 95% of the variation (R^2) in the predictions from the full penalised model (using the method presented in chapter 5.5 of¹⁵). For more details, see the [supplementary materials](#). Our final PBC model contains the predictors age, albumin, cholesterol, and the clinical findings of hepatomegaly and spider nevi. The final model formula is:

$$Pr(\text{fibrosis stage 4}) = \frac{1}{1 + \exp(-X\hat{\beta})},$$

where

$$\begin{aligned} X\hat{\beta} = & 2.42 + 1.48[\text{hepatomegaly}] + 0.579[\text{spiders}] + 0.0198\text{age} \\ & - 0.00249 \text{platelets} - 1.13 \text{albumin} - 0.302 (\text{albumin} - 3.08)_+^3 \\ & + 0.612 (\text{albumin} - 3.57)_+^3 - 0.311 (\text{albumin} - 4.05)_+^3 \\ & - 0.0016 \text{cholesterol} + 1.91 \times 10^{-9} (\text{cholesterol} - 217)_+^3 \\ & - 2.61 \times 10^{-9} (\text{cholesterol} - 310)_+^3 \\ & + 6.98 \times 10^{-9} (\text{cholesterol} - 565)_+^3 \end{aligned}$$

Here, the brackets “[]” equal 1 if the patient has the symptom in question, and 0 otherwise; and braces with a plus “()₊” equal the value inside the braces if that value is positive, and 0 otherwise. The latter only applies to albumin and cholesterol and is a consequence of modelling them using splines.

In the example case of FIB-4, according to the original paper,⁵ the score was developed using a combination of univariate variable selection and subsequent stepwise selection. With only 115 cases of advanced fibrosis to fit the model to, the risk of both eliminating predictive variables and including non-predictive variables is high. If we tried to replicate the model development on another equally sized sample from the same target population, we could end up with a completely different set of predictors. These problematic variable selection techniques do not preclude FIB-4 from being able predict the presence of advanced fibrosis, but they can certainly negatively affect the predictive performance.

5) Generate risk predictions from the model

Once a model has been fitted, it needs to be able to generate risk predictions based on new input. This process should be straight-forward in most cases. In the case of our PBC model, we simply plug in the predictors of new patients into the logistic function, and get a value between 0 and 1, *i.e.* the predicted risk of having cirrhosis. This corresponds to a percentage scale of 0-100% risk. Most software has built-in functions for this.

We would like to stress the importance of having the model generate a probability between 0 and 100%. For example, saying that the risk is 15% tells us more than a score of 4.9 or

“high” ever will. Generating risk scores on the probability scale is also crucial for evaluating the performance of the model (which we shall see in the next step).

In Fig. 2 we have plotted the distributions of predicted risk when we apply our PBC model to the same data, separated by the observed outcome.

FIB-4 is an example of a score which does not represent the probability of an outcome. Instead, a value between 1.30 and 2.66 is generally considered “intermediate risk” and a value of 2.67 or higher is considered “high risk.” These can be used to make medical decisions, but there is no preserved information about what “high risk” means, or what the difference is in risk between patients scoring 1.30, 2.67, or 5.00. The same can be said for a plethora of other widely used risk scores, such as the MELD score and the Child-Pugh score.

6) Evaluate the apparent performance

Now that the model has been defined and fitted, and is capable of generating new predictions, we need to know how our model would perform in practice. This assessment should take three aspects into account: discrimination, calibration, and clinical utility.

a) Discrimination

If a prediction model is to be used for medical decision making, then it should be able to reliably separate those who experience the outcome from those who do not. In other words, a model should be able to discriminate between outcomes based on its predicted risks. Statistically, we want the predicted risks to be higher among the persons with an event than among the persons without an event, so that medical decisions based on the risk model are more likely to be correct. This is best measured with the concordance statistic (c-statistic or c-index),²⁹ which is interpreted as the probability that a randomly selected patient in our data who experiences the event receives a higher predicted risk from our model than a randomly selected patient who does not experience the event.

Graphically, the discrimination can be plotted using a receiver-operating characteristic (ROC) curve. It plots the specificity against the sensitivity of the prediction model at each possible risk threshold. The ROC curve is very often seen,

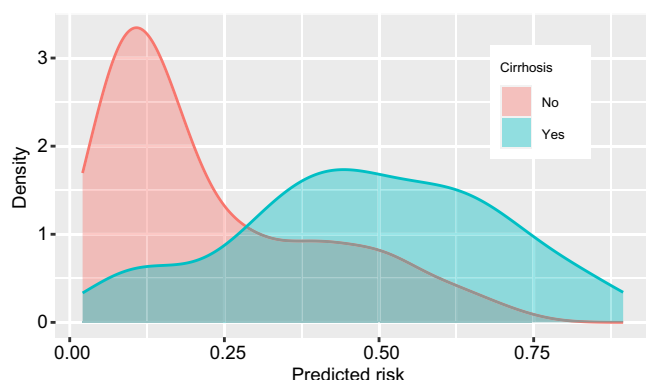


Fig. 2. The distribution of the predicted risks of having cirrhosis from the primary biliary cholangitis model when applied to the same data, separated by the observed outcome. Those without cirrhosis should have a low predicted risk, and those with cirrhosis should have a high predicted risk.

and sometimes attempts are made to find an “optimal” trade-off between sensitivity or specificity. However, the ROC curve obscures important context needed when deciding on a decision threshold: It does not consider the clinical importance of finding true positives relative to avoiding false positives. It also does not tell us which threshold corresponds to which point on the ROC curve. We therefore recommend against analysing ROC curves to find decision thresholds. That is best done using a different technique described below.

It is worth noting that in a diagnostic model (such as in our PBC model), the area under the ROC curve (AUROC or AUC) is equal to the c-index. Quite often, the AUC is reported rather than the c-index. However, in prognostic models, the equivalents to the c-index³⁰ and the time-dependent AUC³¹ are not equal, and the latter is preferred.³² We therefore prefer to use the term AUC rather than c-index for both diagnostic and prognostic models.

In Table 3 we can see that the estimated apparent AUC of our PBC model is 0.81 (95% CI 0.79–0.89). We ignore the ‘corrected’ value for now. We choose not to plot a ROC curve for our model. Any information it can provide other than its AUC can be better studied using a different technique we will describe below (see ‘Clinical utility’).

b) Calibration

In a good prediction model, which generates an absolute risk for an individual, we would like the predicted risk to be close to the actual observed risk. For example, among the group with a predicted risk of around 10%, the observed proportion who experience the event should also be around 10%. This is the calibration of the model.

The best way to assess the calibration is graphically through a calibration plot. This is done by plotting the model-predicted risks on the x-axis, and the corresponding data-observed risks on the y-axis. Perfect calibration would then produce a straight diagonal line. If the outcome is binary, the observed risk needs to be estimated, since we can only observe an event occurring or not. This is best done by fitting a smooth *loess*-based curve from the predicted risks and the corresponding binary outcomes.^{33,34} A diagonal line is plotted as a reference which indicates when the predicted risks equal the observed risks. When the calibration curve is above the diagonal, the predicted risks are too low, and when it is below the diagonal the risks are too high. A typical sign of overfitting in a model is when the curve starts above the diagonal and crosses over to below the diagonal. This indicates that the predictions are too extreme.

We can see the calibration plot for our PBC model in Fig. 3. We ignore the “Corrected” curve for now. Comparing the

Table 3. Discrimination and calibration measures for the primary biliary cholangitis model.

	Apparent	2.5%	97.5%	Corrected
AUC	0.813	0.792	0.893	0.780
ICI	0.023	0.020	0.050	0.013
E50	0.020	0.015	0.048	0.011
E90	0.035	0.034	0.098	0.027
Intercept	0.112	0.077	0.203	-0.022
Slope	1.171	1.166	1.344	0.950

AUC, area under the curve; ICI, integrated calibration index.

The apparent values are given with 95% CIs.

“Apparent” curve to the dotted diagonal line, we see that the predicted risks below 0.27 are slightly overestimated, while predicted risks above 0.27 are underestimated. This pattern can be called underfitting, as the predictions are not extreme enough. This ties back to Step 4, where we intentionally shrunk the model estimates to avoid overfitting.

We highly encourage researchers to include the calibration plot in the main manuscript. Too often it is either omitted entirely or relegated to the [supplementary materials](#).

In addition to the plot, we recommend reporting three measures:³⁵ the integrated calibration index (ICI), which is the mean of the absolute differences between the (smoothed) observed risks and the predicted risks ($\frac{1}{N} \sum |observed_i - predicted_i|$); and the E50 and E90, which are the 50th and 90th percentiles of the absolute differences, respectively. Together, they describe how close our predictions are to the observed outcomes. For example, 90% of individuals will have an absolute prediction error of less than E90.

Other calibration measures one can encounter are the observed:expected (O:E) ratio, the calibration intercept, and the calibration slope. The O:E ratio is calculated by summing up all the predicted risks, which gives the expected number of events (E), and dividing the number of observed events (O) by the expected number. The 95% CI for this ratio is given by

$$\frac{O}{E} \pm 1.96 \sqrt{1/O - 1/N},$$

where N is the total sample size. An O:E ratio of 1 indicates that the average predicted risk corresponds to the proportion of observed events. This is sometimes referred to as calibration-in-the-large.

The calibration intercept and slope are calculated by fitting a logistic regression model with the observed binary outcome,

and with the prediction model’s logit-transformed predictions $\log(\hat{p}/(1-\hat{p}))$ as the only predictor. Such a model produces two estimates: a constant intercept and a regression coefficient for the predictions (the odds ratio in a logistic regression model). This is equivalent to fitting a straight line through the predicted and observed risks in the calibration plot. If there is no systematic over- or underestimation of the risks and there is no over- or underfitting, the estimated intercept will be 0 and the estimated slope coefficient will be 1 (same as the diagonal line in the calibration plot). We have indication of overfitting if the slope is less than 1, and we have an indication of underfitting if the slope is greater than 1.

Generally, the calibration plot and the three measures (ICI, E50, and E90) are more informative of the calibration performance than simply reporting the O:E ratio or calibration intercept and slope.³⁶

The Hosmer-Lemeshow test has been historically used to test if a calibration is in some sense satisfactory.³⁷ However, we discourage using it, as the null hypothesis it tests is akin to “there is no miscalibration”, *i.e.* we would conclude that the model is “well calibrated” if we fail to reject the null hypothesis. It is also generally criticised for lacking statistical power, which makes this null hypothesis especially difficult to reject. For the same reasons we do not recommend setting up a statistical test for whether or not the calibration intercept and slope are 0 and 1, respectively. Absence of evidence is not evidence of absence.³⁸

In [Table 3](#), we can see the estimated ICI, E50, and E90 in our PBC model, with 95% CIs. On average, the prediction error was 2.3 percentage points (95% CI 2.0-5.0), and 90% of patients had a prediction error less than 3.5 percentage points (95% CI 3.4-9.8).

c) Clinical utility

The discrimination and calibration measures give a good insight into how well a model can predict the outcome. However, if the prediction model is to be used, these performance measures need to be tied back to clinical usefulness in some way. At the end of the day, a decision needs to be made for patients based on their predicted risk. This means that some threshold or cut-off for risk needs to be used. A statistical tool for helping to make such a decision is called net benefit analysis.^{39,40} In net benefit we always refer to using a “test”, which is the prediction model used for deciding on a “treatment”, which is the intervention we wish to apply to those with high risk. As previously mentioned, ‘treatment’ in this context can refer to a direct medical treatment or to any other type of intervention, such as further testing (biopsy in the case of our PBC example). Then, for some risk threshold p between 0-1 (0-100%), we treat those who test at or above that threshold and do not treat those below the threshold.

We can always assume that there is a benefit to correctly treating those with the outcome – so-called true positives – and that there is some harm to unnecessarily treating those without the outcome – false positives. For example, if the treatment is a blood test, the harm is negligible except for the time and effort required to collect and analyse it. If the treatment is instead a biopsy, the harm is more time and effort, plus discomfort, pain, or worse for the patient. The different risks of harm will likely

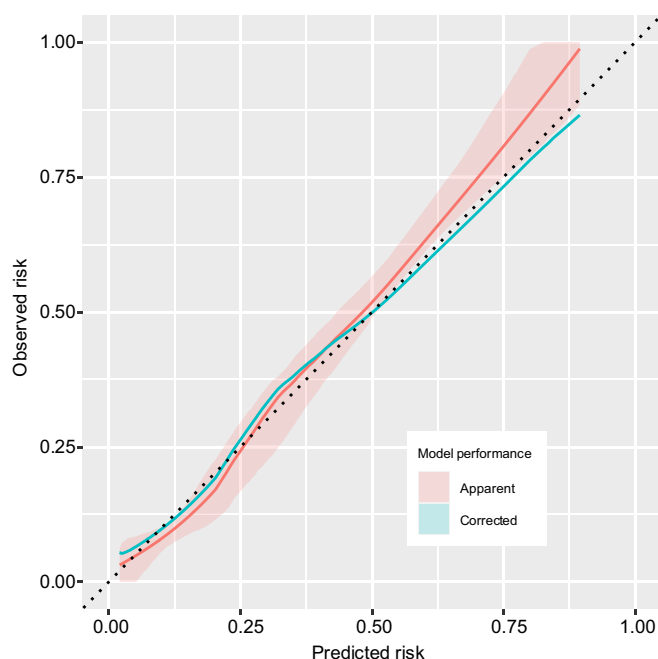


Fig. 3. Calibration curves for the primary biliary cholangitis model. The dotted line indicates correspondence between predicted and observed risks.

cause us to use different risk thresholds for the blood test and the biopsy.

The number of true positives and false positives the prediction model creates for us depends on the chosen risk threshold p , since only those above the threshold receive the treatment. The total benefit of treating the resulting number of true positives is then compared against the total harm of treating the resulting number of false positives. We do this with the predictive net benefit, which for a given threshold p is defined as $NB(p) = \frac{TP}{N} - \frac{p}{1-p} \frac{FP}{N}$, where TP and FP are the number of observed true positives and false positives among those above the threshold p , and N is the total number of patients in the dataset.

The threshold probability we choose determines the acceptable trade-off between benefit and harm. For example, among the patients with a predicted risk of 20% we only expect one in five to experience the event. If we treat those patients, we expect there to be four false positives for every true positive. If we choose 20% as the risk threshold, we imply that we consider the benefit of a necessary treatment to be 4 times greater than the harm of one unnecessary treatment, since the treatment decision is expected to only be correct for one in five patients. In other words, for each correctly treated case we are willing to treat five people (one correctly and four unnecessarily). This is what the formula for net benefit represents.

Net benefit is presented by plotting decision curves. For a reasonable range of thresholds, we plot the threshold on the x-axis and the corresponding net benefit of our prediction model on the y-axis. The plot is then supplemented with two default treatment strategies: “Treat none” and “Treat all.” The “Treat none” strategy classifies no one as positive and always has a net benefit of 0 for all thresholds, since no one is ever selected for treatment. In the “Treat all” strategy everyone is treated no matter the threshold, and so the threshold simply determines the benefit-to-harm ratio of treating everyone. It is also possible to use decision curves to compare multiple prediction models (as long as they aim to predict the same outcome). For example, we can compare our proposed model to a currently established model.

A decision curve for our PBC model can be seen in Fig. 4. The dotted line represents “Treat none”, and the dashed line represents “Treat all.” In this example, “Treat all” means that we suspect everyone of having cirrhosis and that we will refer everyone for biopsy to confirm, while “Treat none” means no one is suspected of cirrhosis and no one is biopsied. The apparent net benefit of the model and “Treat all” is presented with 95% confidence regions. If we believe that 10% is an appropriate threshold, we can read that the net benefit is 0.19 (95% CI 0.14–0.25), which is higher than the net benefit of the “Treat all” strategy. We can also calculate the net benefit using the formula:

$$NB(0.25) = \frac{TP}{N} - \frac{0.25}{1-0.25} \frac{FP}{N} = \frac{77}{288} - \frac{0.25}{0.75} \frac{70}{288} = 0.19,$$

where we had 77 true positives and 70 false positives among those with a predicted risk greater than or equal to 25%.

Perhaps counterintuitively, the decision curve is not meant to be used to find a threshold. Furthermore, there is no

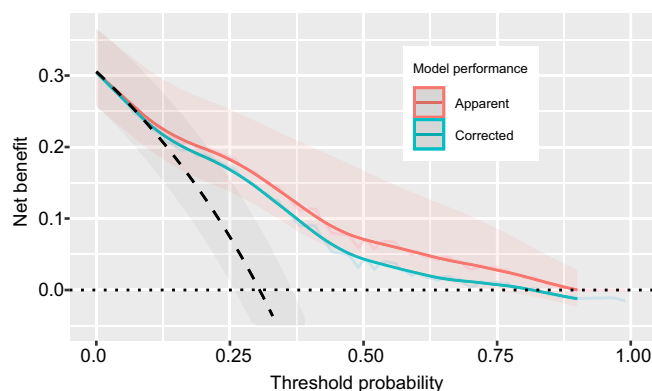


Fig. 4. Decision curves for our primary biliary cholangitis model. The dotted line represents “Treat none”, and the dashed line represents “Treat all” (with 95% confidence regions). As we move from left to right on the threshold probability axis, we are less willing to treat those without the disease in order to correctly treat those with the disease. For example, when our threshold probability is 0.40, we are willing to treat the patient if her predicted risk of having the disease is at least 40%. Then the 40% probability of benefiting from treatment is worth the 60% risk of being harmed by unnecessary treatment.

numerically “optimal” threshold. Instead, a relevant range of thresholds are chosen by the reader based on the perceived ratio between benefit and harm of treatment, and the decision curves are then used to determine which strategy has the highest net benefit at those thresholds. The risk threshold is meant to include every aspect of the benefits and harms depending on who is considering it, be it clinicians, patients, economists, or policy makers. A relatively low threshold implies that it is more important to correctly treat someone with the outcome than it is to not unnecessarily treat someone without the outcome – and vice versa for a relatively high threshold.

7) Internal validation

In the previous step we evaluated the apparent performance of predictions based on the same data with which we developed the prediction model. But in future practice, we will apply the prediction model to new patients not in the original data. Due to overfitting, our prediction model's performance will not be as good in new patients as it was in the patients we used to develop it. We say that the apparent model performance is optimistic, and the degree of optimism follows the degree of overfitting during model development. By internally validating the model, we can estimate this optimism and correct the performance measures to get a more valid estimate of the performance if applied to other patients in the target population.

The most basic way of performing internal validation is by randomly splitting the data into a training dataset on which the model is fitted, and a test dataset for which the predictions are generated. This is meant to emulate applying the prediction model onto new patients, and the estimated performance measures are meant to be more representative of actual performance. The major downside with data splitting is that only a fraction of the data is used for fitting the model (e.g. 70% for training and 30% for testing). Therefore, the overfitting will be

even worse than when using the whole sample. Both the training and testing samples need to be sufficiently large for this type of internal validation to be meaningful.³⁵ The way the splitting is done is also completely arbitrary. Repeating the random splitting will yield slightly different results each time.

Instead, we recommend using bootstrap resampling for internal validation.⁴¹ Bootstrap resampling is a neat trick which gets around having only one sample by treating the data sample as being equivalent to its target population, *i.e.* that a new random sample taken from the data sample is equivalent to a random sample from the target population. If we accept this assumption, then we can resample as much as we want without needing to collect new data. We create a new “bootstrap sample” by randomly drawing – with replacement – patients from the original dataset until the new sample is of the same size as the original dataset. This bootstrap sample will be different from the original due to the random draw (since some patients are drawn multiple times and some are not drawn at all) but will on average have the same features (such as variable means and number of events) as the original. This sampling is repeated hundreds of times. These bootstrap samples can then be used to estimate the uncertainty.

To validate a prediction model using the bootstrap, we fit the prediction model to each bootstrap sample drawn from our original data (we may for example use 500 such samples). Each fitted model is then used to predict the outcome separately in both the bootstrap sample and the original data. Based on those sets of predictions, the performance measures are calculated. This way, we can get an estimate of the difference in performance between the model being applied to its own training data (the apparent performance) and the same model being applied to its target population (the original sample). The average difference over all 500 bootstrap samples is an estimate of the optimism. This estimated optimism is then subtracted from the apparent performance measure to get the optimism-corrected performance measure.

A convenient advantage with using bootstrap for internal validation is that we can get bootstrap confidence intervals for the performance measures at the same time. All confidence intervals and confidence regions presented in our PBC example have come from bootstrapping.

If we revisit Table 3, Fig. 3, and Fig. 4, we can see the optimism-corrected estimates of each measure (referred to as ‘corrected’), based on bootstrap validation with 500 samples. In Table 3, we see that the corrected AUC is slightly lower, losing 0.03 compared to the apparent AUC. The calibration curve in Fig. 3 shows a better calibration when we correct for the optimism. This is because of the penalisation we used in Step 4. We shrunk the model estimates specifically to improve the calibration outside the training sample. The calibration measures in Table 3 are also lower than their apparent counterparts as a result. For the decision curve in Fig. 4, we see a lower net benefit when correcting for the optimism.

8) External validation

In the previous step, we adjusted our measures to get better estimates of the model’s performance in practice. We could provide stronger evidence for real-world performance by validating the model against a completely independent dataset. This is referred to as external validation.

External validation is done simply by using the model developed on the original dataset to generate predictions for the external dataset. The performance measures we get are then estimates of the actual performance on new patients. In practice, there will be some differences in the sample populations between the original training sample and the external validation sample. For example, if a prediction model is trained on all patients with MASLD up to the current year in one country, one might look to validate the model on patients with MASLD from a different country. Since we are estimating the performance measures, the sample size of the validation set also matters for how accurate they are. Some simulation studies have suggested effective sample sizes of at least 100 (*i.e.* cases or events),⁴² while a recent study⁴³ suggests that some settings require many times larger effective sample sizes.

In lieu of a completely external validation set, one can use so-called internal-external validation by splitting the data non-randomly according to some variable which is not a predictor in the model. Calendar time is a good candidate, where patients recruited early into the study are used as the training set, and patients recruited later make up the test set. This temporal validation simulates how the model should perform in the future based on the current data, although this approach can be vulnerable to changes in clinical practice that may have occurred over time. Another choice can be different treatment centres or different regions if there are multiple in the dataset. This would simulate transporting the model to other and (slightly) different populations.

However, internal-external validation will only weakly validate these aspects, since both the training and test sets are from the same study and sampling (and healthcare system). It also has the same problem that random data splitting has regarding producing smaller sample sizes for the training and test sets.

Some aspects to consider when determining if a model is sufficiently validated are the estimated performance measures in the validation datasets, the consistency of these estimates between different validation datasets (including those from internal validation) and the sample size of the validation datasets.

Depending on the differences in populations between the development data and the validation data, we can make different assertions about the generalisability/transportability of the prediction model.⁴⁴ One can use external validation to assess three different types of generalisability:⁴⁵

1. *Temporal generalisation*: Indicates how the prediction model will perform over time in the same target population that it was developed in. For example, how well can a predictive model for advanced fibrosis developed on historic data be expected to perform in current and future patients if the prevalence of type 2 diabetes has since risen in that population?
2. *Geographical generalisation*: Indicates how well the prediction model will perform in a new location – outside the location where the model was developed. For example, if a model is developed based on patient data from one hospital or healthcare region, but it is intended to be implemented nationally (or even internationally).
3. *Domain generalisation*: Indicates how well the model will perform in a new clinical context. For example, FIB-4 was developed for patients with HIV and hepatitis C co-infection, but has been applied to patients with MASLD among others. As

another example, the MELD score was originally intended for TIPS recipients, but was later generalised to all patients with end-stage cirrhosis.

9) Publish the model

The final step is to publish the prediction model. By that, we mean in such a way that it can be used immediately by anyone who wishes to. The formula for the model should be presented in the manuscript. Attempting to hide what the prediction model actually is or does should be avoided. Any potential user of the model needs to be able to trust its contents. For even greater transparency, we recommend publishing the code used to develop the model, either as [supplementary material](#), or externally at an online repository.

A popular way to promote a model is through an online calculator, where the user can input the values of the required predictors and receive predicted risk outputs. A good example is the VOCAL-Penn score for post-operative mortality in patients with cirrhosis (available at vocalpennscore.com). A calculator allows us to design a user-friendly interface even when the model is complex and difficult to implement by oneself. Once again, we stress the need to be transparent about what the underlying model is. An older pen-and-paper approach is the nomogram.⁴⁶

For the PBC model, we have already provided the model formula in Step 4. As a substitute for a web calculator, we have created a nomogram for the model ([Fig. 5](#)). For example, a patient with hepatomegaly (38 points), albumin level of 3.4 g/dl (50 points), platelet count of $200 \times 10^9/L$ (26 points), cholesterol level of 1,200 mg/dl (21 points), no spider nevi (0 points), and age 50 (13 points) will have a total point count of 148. This corresponds to a predicted risk of approximately 25%.

The model derivation is available in the [supplementary materials](#), and the code is available online at <https://github.com/rickstra/PBCexample>.

Additional considerations for prognostic models

When we want to develop and validate a prognostic model with time-to-event data, the general procedure of the nine steps is the same, but with some adjustments in the specifics. Firstly, the outcome of interest now needs to have a fixed time horizon, e.g. 30-day risk or 5-year risk. The predicted risks will be specific to that time horizon and will not be calibrated to any other.

In a prognostic model, the predicted risk for an individual is equivalent to their estimated cumulative incidence function at the time horizon. It can be difficult to produce this function depending on the type of model used in Step 4. If one uses a semi-parametric model such as cause-specific Cox models or the Fine-Gray subdistribution hazard model, the baseline hazard needs to be separately estimated. Most statistical software has a standard method for inferring a baseline hazard, but that method is not always communicated well. It is important to understand how the baseline hazard is estimated, since it is an important component of the prediction model.

The reason we recommend the flexible parametric model over the standard semi-parametric models is that the baseline hazard is modelled jointly with the predictors. The model's formula therefore explicitly tells us the entire hazard function over time. In the presence of competing risks, we need to fit two models – one for the event of interest, and one for the competing event. This makes the approach different from the Fine-Gray model, which regresses on the cumulative incidence

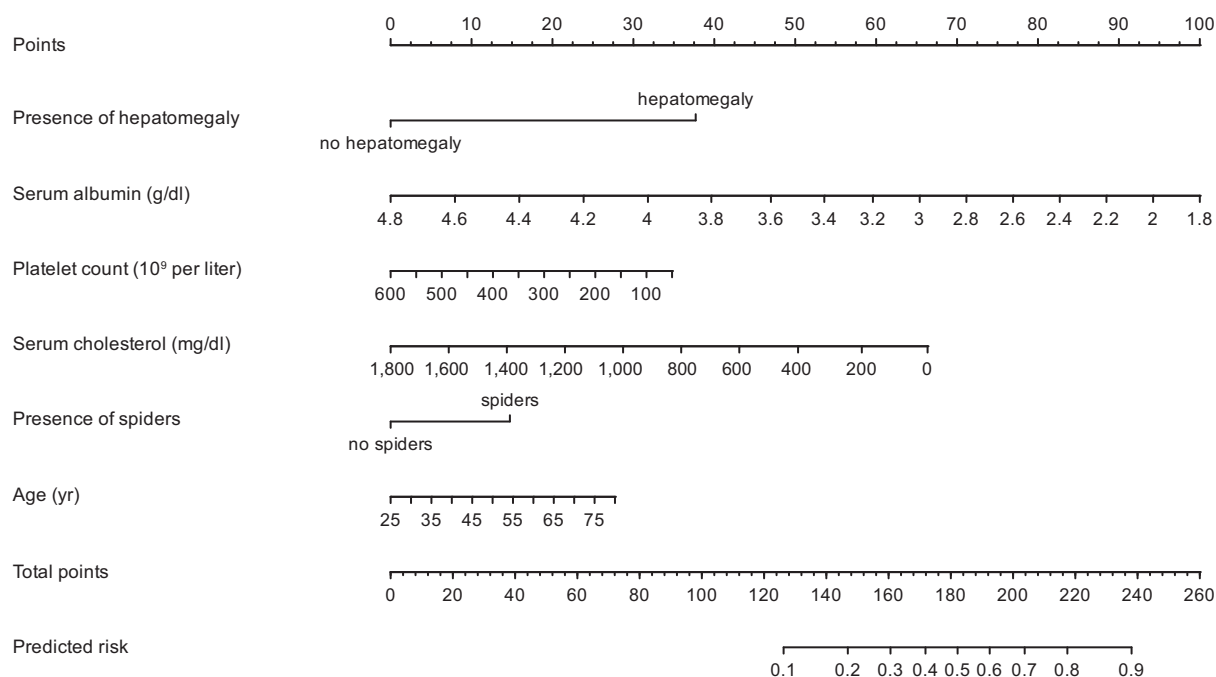


Fig. 5. Nomogram of our primary biliary cholangitis model. Read from the scale of each predictor and determine the corresponding number of points from the top scale. Sum up all the points and read the predicted risk from the bottom scale.

(accounting for competing risks) of the event of interest directly. For a comparative example and discussion of the issues of the Fine-Gray model, see e.g. the vignette for the *survival* package in R.⁴⁷

When assessing the discrimination of a prognostic model at a time horizon t , we need to determine what constitutes experiencing and not experiencing the event. The distinction is clear for those who experience the event before t , and for those who are event-free and not lost to follow-up by time t . But what about those who experience a competing event or are censored before t ? The time-dependent AUC³¹ uses a technique (inverse probability of censoring weighting) to account for this.⁴⁸

When calibrating a prognostic model, the observed risks need to account for censoring and competing risks. The simplest way to calculate the observed risks is to order the predicted risks and then stratify them into equal size groups (e.g. 10 groups corresponding to the 0-9th, 10-19th, 20-29th, ... percentiles of predicted risk) and take the Aalen-Johansen estimates of the cumulative incidence for each group as an estimate of the observed risk. One can then plot the average predicted risk in each group against the respective estimate of observed risk. The disadvantage of this approach is that the plot is sensitive to the method of stratification. An alternative approach has been introduced by Austin *et al.*,⁴⁹ which involves fitting a Fine-Gray model with a restricted cubic spline function of the risk predictions as the only predictor. The fitted values of this model then constitute the observed risks (similar to the *loess*-smoothed values in the diagnostic calibration). From this, we can also get the ICI, E50 and E90.

When we calculate the net benefit of a prognostic model at a threshold, we can no longer simply count the number of true positives and false positives, again owing to censoring and competing risks. The number of true positives at a threshold is instead the estimated cumulative incidence among those with predicted risks at or above the threshold multiplied by the total number at or above the threshold. The number of false

positives is calculated analogously using one minus the cumulative incidence.⁴⁰

Final remarks

We have presented nine steps to develop and validate a clinical prediction model. There are more details to consider at each step, but we have provided the basic concepts and the reasoning behind them.

The main assessment we have emphasised is net benefit and decision curves. These incorporate both discrimination and calibration performance, and relate them more closely to clinical decision making. Indeed, poor discrimination or calibration will reduce the net benefit.⁵⁰ The net benefit depends on a risk threshold, which we need to specify ourselves based on our own assessment of benefit vs. harm. This is in contrast to many proposed models which immediately try to define an optimal cut-off. We argue that this should wait and that it cannot be determined using only these statistical tools. The next step for a prediction model which shows promise over a wide range of thresholds would be to perform a proper cost-benefit analysis, and to involve more experts and patient groups to determine the appropriate threshold based on all available information in the field.

In 2015, the TRIPOD initiative (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) released guidelines for how to report the development and validation of clinical prediction models.^{51,52} Included in the guidelines is a handy checklist which we recommend for use while writing manuscripts with prediction models, as well as when reviewing manuscripts with prediction models.

Our hope is that the steps we have outlined will help researchers in hepatology to both develop and validate new prediction models, as well as to better assess the quality of other researchers' pre-existing or proposed prediction models. Higher quality in both these aspects would significantly improve research and lead to better decision making.

Affiliations

¹Department of Medicine, Huddinge, Karolinska Institutet, Stockholm, Sweden; ²Department of Hepatology and Gastroenterology, Aarhus University Hospital, Aarhus, Denmark; ³Department of Clinical Epidemiology, Aarhus University Hospital, Aarhus, Denmark; ⁴Division of Hepatology, Department of Upper GI, Karolinska University Hospital, Stockholm, Sweden

Abbreviations

AUC, area under the curve; ICI, integrated calibration index; MASLD, metabolic dysfunction-associated steatotic liver disease; MELD, model for end-stage liver disease; NB, net benefit; PBC, primary biliary cholangitis; ROC, receiver-operating characteristic; TIPS, transjugular intrahepatic portosystemic shunt.

Financial support

The authors did not receive any financial support to produce this manuscript.

Conflicts of interest

PJ's research is supported by a grant from the Novo Nordisk Foundation (NNF19OC0054612). HH's institutions have received research funding from Astra Zeneca, EchoSens, Gilead, Intercept, MSD, Novo Nordisk and Pfizer. He has served as consultant or on advisory boards for Astra Zeneca, Bristol Myers-Squibb, MSD and Novo Nordisk and has been part of hepatic events adjudication committees for Boehringer Ingelheim, KOWA and GW Pharma. All unrelated to the current manuscript.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

Study conception and design: All. Acquisition of data: N/A. Statistical analysis: RS. Analysis and interpretation of data: RS. Drafting of manuscript: RS. Critical revision: PJ, HH. Guarantor of article: RS. All authors approved the final version of the article, including the authorship list.

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhep.2024.03.030>.

References

- [1] Hennes EM, Zeniya M, Czaja AJ, et al. Simplified criteria for the diagnosis of autoimmune hepatitis. *Hepatology* 2008;48:169–176. <https://doi.org/10.1002/hep.22322>.
- [2] Malinchoc M, Kamath PS, Gordon FD, et al. A model to predict poor survival in patients undergoing transjugular intrahepatic portosystemic shunts. *Hepatology* 2000;31:864–871. <https://doi.org/10.1053/he.2000.5852>.

- [3] Hernán MA, Hsu J, Healy B. A second chance to get causal inference right: a classification of data science tasks. *Chance* 2019;32:42–49. <https://doi.org/10.1080/09332480.2019.1579578>.
- [4] Anstee QM, Castera L, Loomba R. Impact of non-invasive biomarkers on hepatology practice: past, present and future. *J Hepatol* 2022;76:1362–1378. <https://doi.org/10.1016/j.jhep.2022.03.026>.
- [5] Sterling RK, Lissen E, Clumeck N, et al. Development of a simple noninvasive index to predict significant fibrosis in patients with HIV/HCV coinfection. *Hepatology* 2006;43:1317–1325. <https://doi.org/10.1002/hep.21178>.
- [6] Therneau TM, Grambsch PM. Modeling survival data: extending the Cox model. New York, NY: Springer New York; 2000. <https://doi.org/10.1007/978-1-4757-3294-8>.
- [7] Donders ART, van der Heijden GJMG, Stijnen T, et al. Review: a gentle introduction to imputation of missing values. *J Clin Epidemiol* 2006;59:1087–1091. <https://doi.org/10.1016/j.jclinepi.2006.01.014>.
- [8] Altman DG, Vergouwe Y, Royston P, et al. Prognosis and prognostic research: validating a prognostic model. *BMJ (Online)* 2009;338:1432–1435. <https://doi.org/10.1136/bmj.b605>.
- [9] Kamath PS, Wiesner RH, Malinchoc M, et al. A model to predict survival in patients with end-stage liver disease. *Hepatology* 2001;33:464–470. <https://doi.org/10.1053/jhep.2001.22172>.
- [10] Kim WR, Biggins SW, Kremers WK, et al. Hyponatremia and mortality among patients on the liver-transplant waiting list. *New Engl J Med* 2008;359:1018–1026. <https://doi.org/10.1056/nejmoa0801209>.
- [11] Kim WR, Mannalithara A, Heimbach JK, et al. MELD 3.0: the model for end-stage liver disease updated for the modern era. *Gastroenterology* 2021;161:1887–1895.e4. <https://doi.org/10.1053/j.gastro.2021.08.050>.
- [12] Schattenberg JM, Anstee QM, Caussy C, et al. Differences between current clinical guidelines for screening, diagnosis and management of nonalcoholic fatty liver disease and real-world practice: a targeted literature review. *Expert Rev Gastroenterol Hepatol* 2021;15:1253–1266. <https://doi.org/10.1080/17474124.2021.1974295>.
- [13] Wyatt JC, Altman DG. Commentary: prognostic models: clinically useful or quickly forgotten? *BMJ* 1995;311:1539–1541. <https://doi.org/10.1136/bmj.311.7019.1539>.
- [14] Steyerberg EW, Eijkemans MJC, Harrell FE, et al. Prognostic modeling with logistic regression analysis: in search of a sensible strategy in small data sets. *Med Decis Making* 2001;21:45–56. <https://doi.org/10.1177/0272989X0102100106>.
- [15] Harrell FE. Regression modeling strategies, 13. Cham: Springer International Publishing; 2015. <https://doi.org/10.1007/978-3-319-19425-7>.
- [16] Peduzzi P, Concato J, Kemper E, et al. A simulation study of the number of events per variable in logistic regression analysis. *J Clin Epidemiol* 1996;49:1373–1379. [https://doi.org/10.1016/S0895-4356\(96\)00236-3](https://doi.org/10.1016/S0895-4356(96)00236-3).
- [17] Riley RD, Ensor J, Snell KIE, et al. Calculating the sample size required for developing a clinical prediction model. *The BMJ* 2020;368:1–12. <https://doi.org/10.1136/bmj.m441>.
- [18] Van Houwelingen JC, Le Cessie S. Predictive value of statistical models. *Stat Med* 1990;9:1303–1325. <https://doi.org/10.1002/sim.4780091109>.
- [19] White IR, Royston P, Wood AM. Multiple imputation using chained equations: issues and guidance for practice. *Stat Med* 2011;30:377–399. <https://doi.org/10.1002/sim.4067>.
- [20] Knol MJ, Janssen KJM, Donders ART, et al. Unpredictable bias when using the missing indicator method or complete case analysis for missing confounder values: an empirical example. *J Clin Epidemiol* 2010;63:728–736. <https://doi.org/10.1016/j.jclinepi.2009.08.028>.
- [21] Fine JP, Gray RJ. A proportional hazards model for the subdistribution of a competing risk. *J Am Stat Assoc* 1999;94:496–509. <https://doi.org/10.1080/01621459.1999.10474144>.
- [22] Royston P, Parmar MKB. Flexible parametric proportional-hazards and proportional-odds models for censored survival data, with application to prognostic modelling and estimation of treatment effects. *Stat Med* 2002;21:2175–2197. <https://doi.org/10.1002/sim.1203>.
- [23] Lambert PC, Wilkes SR, Crowther MJ. Flexible parametric modelling of the cause-specific cumulative incidence function. *Stat Med* 2017;36:1429–1446. <https://doi.org/10.1002/sim.7208>.
- [24] Royston P, Altman DG, Sauerbrei W. Dichotomizing continuous predictors in multiple regression: a bad idea. *Stat Med* 2006;25:127–141. <https://doi.org/10.1002/sim.2331>.
- [25] Royston P, Sauerbrei W. Multivariable model-building. Wiley; 2008. <https://doi.org/10.1002/9780470770771>.
- [26] Moons KGM, Donders ART, Steyerberg EW, et al. Penalized maximum likelihood estimation to directly adjust diagnostic and prognostic prediction models for overoptimism: a clinical example. *J Clin Epidemiol* 2004;57:1262–1270. <https://doi.org/10.1016/j.jclinepi.2004.01.020>.
- [27] Tibshirani R. The lasso method for variable selection in the cox model. *Stat Med* 1997;16:385–395. [https://doi.org/10.1002/\(SICI\)1097-0258\(19970228\)16:4<385::AID-SIM380>3.0.CO;2-395](https://doi.org/10.1002/(SICI)1097-0258(19970228)16:4<385::AID-SIM380>3.0.CO;2-395).
- [28] Zhang HH, Lu W. Adaptive Lasso for Cox's proportional hazards model. *Biometrika* 2007;94:691–703. <https://doi.org/10.1093/biomet/asm037>.
- [29] Harrell FE, Califf RM, Pryor DB, et al. Evaluating the yield of medical tests. *JAMA: J Am Med Assoc* 1982;247:2543–2546. <https://doi.org/10.1001/jama.1982.03320430047030>.
- [30] Wolbers M, Blanche P, Koller MT, et al. Concordance for prognostic models with competing risks. *Biostatistics* 2014;15:526–539. <https://doi.org/10.1093/biostatistics/kxt059>.
- [31] Blanche P, Dartigues J-F, Jacqmin-Gadda H. Estimating and comparing time-dependent areas under receiver operating characteristic curves for censored event times with competing risks. *Stat Med* 2013;32:5381–5397. <https://doi.org/10.1002/sim.5958>.
- [32] Blanche P, Kattan MW, Gerds TA. The c-index is not proper for the evaluation of t-year predicted risks. *Biostatistics* 2019;20:347–357. <https://doi.org/10.1093/biostatistics/kxy006>.
- [33] Austin PC, Steyerberg EW. Graphical assessment of internal and external calibration of logistic regression models by using loess smoothers. *Stat Med* 2014;33:517–535. <https://doi.org/10.1002/sim.5941>.
- [34] Austin PC, Harrell FE, Van Klaveren D. Graphical calibration curves and the integrated calibration index (ICI) for survival models. *Stat Med* 2020;39:2714–2742. <https://doi.org/10.1002/sim.8570>.
- [35] Austin PC, Steyerberg EW. The Integrated Calibration Index (ICI) and related metrics for quantifying the calibration of logistic regression models. *Stat Med* 2019;38:4051–4065. <https://doi.org/10.1002/sim.8281>.
- [36] Van Calster B, Nieboer D, Vergouwe Y, et al. A calibration hierarchy for risk models was defined: from utopia to empirical data. *J Clin Epidemiol* 2016;74:167–176. <https://doi.org/10.1016/j.jclinepi.2015.12.005>.
- [37] Hosmer DW, Lemeshow S, Sturdivant RX. Applied logistic regression, 85. Wiley; 2013. <https://doi.org/10.1002/9781118548387>.
- [38] Altman DG, Bland JM. Statistics notes: absence of evidence is not evidence of absence. *BMJ* 1995;311:485. <https://doi.org/10.1136/bmj.311.7003.485>.
- [39] Vickers AJ, Van Calster B, Steyerberg EW. Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests. *BMJ (Online)* 2016;352:3–7. <https://doi.org/10.1136/bmj.i6>.
- [40] Vickers AJ, Cronin AM, Elkin EB, et al. Extensions to decision curve analysis, a novel method for evaluating diagnostic tests, prediction models and molecular markers. *BMC Med Inform Decis Making* 2008;8:1–17. <https://doi.org/10.1186/1472-6947-8-53>.
- [41] Efron B, Tibshirani RJ. An introduction to the bootstrap. Chapman: Hall/CRC; 1994. <https://doi.org/10.1201/9780429246593>.
- [42] Vergouwe Y, Steyerberg EW, Eijkemans MJC, et al. Substantial effective sample sizes were required for external validation studies of predictive logistic regression models. *J Clin Epidemiol* 2005;58:475–483. <https://doi.org/10.1016/j.jclinepi.2004.06.017>.
- [43] Riley RD, Debray TPA, Collins GS, et al. Minimum sample size for external validation of a clinical prediction model with a binary outcome. *Stat Med* 2021;40:4230–4251. <https://doi.org/10.1002/sim.9025>.
- [44] Justice AC, Covinsky KE, Berlin JA. Assessing the generalizability of prognostic information. *Ann Intern Med* 1999;130:515–524. <https://doi.org/10.7326/0003-4819-130-6-199903160-00016>.
- [45] Hond AAH de, Shah VB, Kant IMJ, et al. Perspectives on validation of clinical predictive algorithms. *Npj Digital Med* 2023;6:86. <https://doi.org/10.1038/s41746-023-00832-9>.
- [46] Banks J. Nomograms. Wiley StatsRef: statistics reference online. Wiley; 2014. p. 1–10. <https://doi.org/10.1002/9781118445112.stat02304>.
- [47] Therneau TM. A package for survival analysis in R. 2023.
- [48] Van Geloven N, Giardiello D, Bonneville EF, et al. Validation of prediction models in the presence of competing risks: a guide through modern methods. *The BMJ* 2022;1–7. <https://doi.org/10.1136/bmj-2021-069249>.

- [49] Austin PC, Putter H, Giardiello D, et al. Graphical calibration curves and the integrated calibration index (ICI) for competing risk models. *Diagn Prognostic Res* 2022;6. <https://doi.org/10.1186/s41512-021-00114-6>.
- [50] Van Calster B, Vickers AJ. Calibration of risk prediction models: impact on decision-analytic performance. *Med Decis Making* 2015;35:162–169. <https://doi.org/10.1177/0272989X14547233>.
- [51] Collins GS, Reitsma JB, Altman DG, et al. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD Statement. *BMC Med* 2015;13:1. <https://doi.org/10.1186/s12916-014-0241-z>.
- [52] Moons KGM, Altman DG, Reitsma JB, et al. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): explanation and elaboration. *Ann Intern Med* 2015;162:W1–W73. <https://doi.org/10.7326/M14-0698>.

Journal of Hepatology, Volume 81

Supplemental information

Developing and validating clinical prediction models in hepatology – An overview for clinicians

Rickard Strandberg, Peter Jepsen, and Hannes Hagström

Developing and Validating Clinical Prediction Models in Hepatology—an Overview for Clinicians

Supplementary Materials

Rickard Strandberg, Peter Jepsen, Hannes Hagström

Table of contents

1. [Developing the PBC Model](#)
2. [Sample Size Calculations](#)
3. [Stepwise Simulation Example](#)

1. Developing the PBC Model

In this section we will go through all the model development steps used in producing the PBC cirrhosis prediction model used as the example in the main manuscript. The steps more or less follow the recommendations in [1]. All code is also available at <https://github.com/rickstra/PBCexample>.

12 potential predictors are chosen after some deliberation (where patients with ascites are excluded from the sample and urine copper and prothrombin time are excluded due to not being feasible in the intended setting). 4 of the predictors (sex, hepatomegaly, spiders, and edema) are categorical, and the remaining 8 are continuous.

As a first step we can consider the most complex model we are willing to consider. This is a model where all 12 predictors are included, all categorical predictors retain all their categories (2 categories for all except edema which has 3), and all continuous predictors are allowed to be non-linear. In this case, where the effective sample size is limited, the most flexibility we are willing to provide is 2 degrees of freedom, which we get by modelling the continuous predictors as 3-knot restricted cubic splines.

With this procedure we give each predictor an equal and generous chance to fit to the data. Based on how well each predictor can explain the data, we can then simplify some of the predictors (i.e. reduce those predictors' degrees of freedom). Note that we will not be using the results to eliminate predictors—all 12 predictors are still included with at least 1 degree of freedom—but rather to allocate degrees of freedom according to the perceived strength of each predictor.

We fit this “full” logistic regression model:

```
## stage4 ~ sex + rcs(age, 3) + rcs(bilirubin, 3) + rcs(albumin,
##      3) + rcs(ast, 3) + rcs(platelets, 3) + rcs(cholesterol, 3) +
##      rcs(alk.phos, 3) + rcs(triglycerides, 3) + spiders + hepato +
##      edema

## Logistic Regression Model
##
## lrm(formula = f_full, data = d)
##
##               Model Likelihood      Discrimination      Rank Discrim.
##               Ratio Test      Indexes      Indexes
## Obs           288    LR chi2    101.95      R2           0.421      C           0.845
## 0             200    d.f.         21      R2(21,288)0.245      Dxy           0.689
## 1             88    Pr(> chi2) <0.0001  R2(21,183.3)0.357      gamma          0.689
## max |deriv| 4e-05      Brier       0.141      tau-a          0.293
```

This model has 21 degrees of freedom total. We know that this is too many by design, so we need to find predictors where we can reduce the number of degrees of freedom.

We plot the chi-squared (χ^2) statistics of each predictor, which is a statistic indicating how well the predictor explains the data (and is used for calculating the p-value of the predictor). The expected value of the statistic if the predictor is truly uninformative is equal

to the number of degrees of freedom used. If we subtract the degrees of freedom from each statistic they all have an expected value of 0. We can then rank the predictors and reduce the degrees of freedom used for the less informative predictors. The results are in Figure S1.

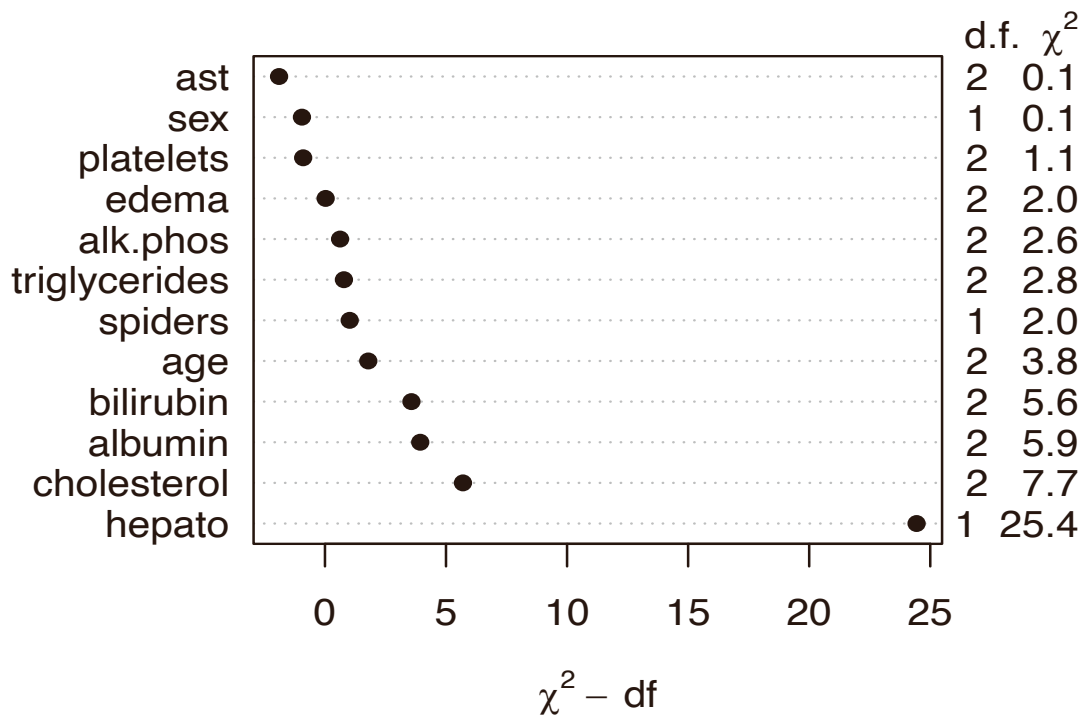


Figure S1: Comparing the chi-squared statistics of each predictor in a “full” model where all categorical predictors include all their categories and all continuous predictors are modelled with 3-knot restricted cubic splines (2 degrees of freedom each).

In Figure S1 we see that the presence of hepatomegaly is the strongest predictor followed by a large gap to cholesterol, and then a smaller gap to albumin and bilirubin, and another gap to the rest of the predictors. Based on this, we make the judgement that cholesterol, albumin, and bilirubin can keep their 2 degrees of freedom since they are the most promising continuous predictors (hepatomegaly only has 1 degree of freedom regardless). The rest of the predictors are reduced to 1 degree of freedom each, meaning that age, triglycerides, alkaline phosphatase, platelets, and AST are made linear and that the 3 categories of edema are reduced to 2: “treated or untreated edema” and “no edema”.

We fit this “reduced” model:

```
## stage4 ~ ast + triglycerides + sex + spiders + edema01 + platelets +
##      age + alk.phos + rcs(bilirubin, 3) + rcs(albumin, 3) + rcs(cholesterol,
##      3) + hepato
```

```
## Logistic Regression Model
##
## lrm(formula = f, data = d, x = T, y = T)
##
##           Model Likelihood      Discrimination      Rank Discrim.
##           Ratio Test           Indexes           Indexes
## Obs          288    LR chi2      97.82          R2          0.407    C          0.837
## 0            200    d.f.          15          R2(15,288)0.250    Dxy          0.674
## 1             88    Pr(> chi2) <0.0001    R2(15,183.3)0.363    gamma          0.674
## max |deriv| 2e-05          Brier      0.145    tau-a          0.287
```

We can then check that the Akaike Information Criterion (AIC) is lower for the reduced model, indicating that the degrees of freedom we removed do not significantly improve the fit, i.e. that they are at least somewhat redundant:

```
##      AIC_full AIC_reduced
##      296.5766   288.7053
```

The 15 degrees of freedom remaining is still too many. We can quickly calculate the heuristic shrinkage factor [2] for the unpenalized model:

$$\frac{\chi^2 - df}{\chi^2} = 0.85$$

which is lower than the 0.9 recommended by the sample size calculations in the main manuscript [3]. We should therefore use some type of shrinkage method.

We use penalized maximum likelihood estimation as our shrinkage method [4]. To select the tuning parameter λ of the penalized regression, we do a grid search over a range of values. We can let the value of λ be different for the simple (linear) effects of the predictors and for the non-linear effects. We find the pair of values which optimize the AIC:

```
##      simple nonlinear
## 3      4.7         23.5
```

We then fit the penalized version of the reduced model using these values of λ :

```
## Logistic Regression Model
##
## lrm(formula = f, data = d, x = T, y = T, penalty = list(simple =
##      nonlinear = opt.pen$nonlinear))
##
## Penalty factors
##
##      simple nonlinear interaction nonlinear.interaction
##      4.7         23.5          23.5          23.5
##
##           Model Likelihood      Discrimination      Rank Discrim.
##           Ratio Test           Indexes           Indexes
## Obs          288    LR chi2      89.37          R2          0.344    C          0.823
```

## 0	200	d.f.	10.359	R2(15,288)0.203	Dxy	0.646
## 1	88	Pr(> chi2)<0.0001		R2(15,183.3)0.300	gamma	0.646
## max deriv	1e-09	Penalty	9.02	Brier	0.149	tau-a 0.275

This penalized model has an effective number of degrees of freedom of 10.36 compared to the 15 of the unpenalized model.

If we can accept a prediction model which has all 12 predictors for clinical use, we are finished with the model. However, to make the PBC model more practical to use clinically we want to reduce the number of predictors we need to measure.

We must first acknowledge that the penalized model with all 12 predictors is the better model for accurately predicting the outcome and has an appropriate amount of shrinkage. We can then investigate if there is a small subset of predictors which can adequately approximate the predictions of the 12 predictor model. To do this, we treat the linear predictors ($logit(\hat{p})$) generated by the 12 predictor model as the outcome in an ordinary linear regression which includes all 12 predictors as they are in the prediction model. The formula is:

```
## lp ~ ast + triglycerides + sex + spiders + edema01+ platelets +
## age + alk.phos + rcs(bilirubin, 3) + rcs(albumin, 3) + rcs(cholesterol,
## 3) + hepato
```

where lp is the linear predictor of the penalized 12 predictor model.

We then perform a backwards elimination procedure where at each step we eliminate the predictor which reduces the R^2 the least, until all predictors are eliminated.

```
##
## Deleted      Chi-Sq d.f. P      Residual d.f. P      AIC      R2
## alk.phos      0.58 1      0.4450      0.58      1      0.4450      -1.42 0.999
## sex           1.37 1      0.2421      1.95      2      0.3768      -2.05 0.996
## ast           1.52 1      0.2181      3.47      3      0.3248      -2.53 0.992
## edema01       1.68 1      0.1952      5.15      4      0.2726      -2.85 0.989
## bilirubin     5.39 2      0.0675     10.54      6      0.1037      -1.46 0.977
## triglycerides 10.22 1      0.0014     20.76      7      0.0041       6.76 0.955
## age          11.21 1      0.0008     31.97      8      0.0001     15.97 0.930
## spiders      14.04 1      0.0002     46.01      9      0.0000     28.01 0.900
## platelets     19.06 1      0.0000     65.07     10      0.0000     45.07 0.858
## cholesterol  43.62 2      0.0000    108.69     12      0.0000     84.69 0.763
## albumin      85.20 2      0.0000    193.89     14      0.0000    165.89 0.577
## hepato       264.29 1      0.0000    458.18     15      0.0000    428.18 0.000
##
## Approximate Estimates after Deleting Factors
##
##      Coef      S.E. Wald Z P
## [1,] -1.084 0.05893 -18.4 0
##
## Factors in Final Model
##
## None
```

The final simplified model we choose is the one with the fewest predictors but with an $R^2 \geq 0.95$ (see last column above), which is the one with hepatomegaly, albumin, cholesterol, platelets, spiders, and age.

The final approximated model we intend for use is then the logistic regression model with only those 6 predictors. Instead of re-fitting the model using (penalized) maximum likelihood as usual, we calculate the predictor coefficients and variance-covariance matrix of the model based on those of the full penalized 12 predictor model. That way, the shrinkage is preserved and the standard errors (and confidence intervals) reflect that the model was approximated from a larger model.

If $\hat{\beta}$ are the estimated predictor coefficients from the penalized 12 predictor model and X is the model's design matrix (i.e. the 12 predictors), and Z is the design matrix for the approximated model (the 6 predictors), then the estimated predictor coefficients of the approximated 6 predictor model is $W\hat{\beta}$, where $W = (Z^T Z)^{-1} Z^T X$. The variance-covariance matrix of the approximated model is $W V W^T$, where V is the variance-covariance of the 12 predictor model.

The final model formula is:

$$Pr(\text{fibrosis stage 4}) = \frac{1}{1 + \exp(-X\hat{\beta})},$$

where

$$\begin{aligned} X\hat{\beta} = & 2.42 + 1.48[\text{hepatomegaly}] + 0.579[\text{spiders}] + 0.0198\text{age} - 0.00249\text{platelets} \\ & - 1.13\text{albumin} - 0.302(\text{albumin} - 3.08)_+^3 + 0.612(\text{albumin} - 3.57)_+^3 \\ & - 0.311(\text{albumin} - 4.05)_+^3 - 0.0016\text{cholesterol} + 1.91 \times 10^{-9}(\text{cholesterol} - 217)_+^3 \\ & - 2.61 \times 10^{-9}(\text{cholesterol} - 310)_+^3 + 6.98 \times 10^{-9}(\text{cholesterol} - 565)_+^3 \end{aligned}$$

Here, the brackets “[]” equal 1 if the patient has the symptom in question, and 0 otherwise; and braces with a plus “()₊” equal the value inside the braces if that value is positive, and 0 otherwise. The latter only applies to albumin and cholesterol and is a consequence of modeling them using splines.

2. Sample Size Calculation

In this section we outline the formulas and calculations needed for determining the minimum sample size needed according to the 4 criteria by Riley et al. [3]. If the data has already been collected and the sample size is known, we can also plug that number into these formulas to calculate the number of degrees of freedom which could be used to satisfy the 4 criteria.

Criterion 1

The first step is to ensure sufficient sample size to accurately estimate the event proportion ϕ such that the 95% confidence interval is within $\phi \pm \delta$ for some margin of error δ . The formula is

$$n_1 = \left(\frac{1.96}{\delta} \right)^2 \phi(1 - \phi)$$

In our PBC data we have an event (cirrhosis) proportion of $\phi = 0.306$. Plugging this into the above formula together with $\delta = 0.05$, we get

$$n_1 = \left(\frac{1.96}{0.05} \right)^2 0.306(1 - 0.306) = 326$$

That should be enough to keep the 95% confidence interval for ϕ approximately within 5 percentage points of 0.306.

Criterion 2

To get the sample size needed to control the mean average prediction error (MAPE), plug in the proportion experiencing the event ϕ , the number of model degrees of freedom p , and the desired MAPE (recommended MAPE=0.05) into the following formula:

$$n_2 = \exp \left(\frac{-0.508 + 0.259 \ln(\phi) + 0.504 \ln(p) - \ln(MAPE)}{0.544} \right)$$

In our PBC data we have an event (cirrhosis) proportion of $\phi = 0.306$. If we plug this into the formula above together with a preliminary $p = 14$ degrees of freedom and a desired MAPE of 0.05, we get

$$n_2 = \exp \left(\frac{-0.508 + 0.259 \ln(0.306) + 0.504 \ln(14) - \ln(0.05)}{0.544} \right) = 635.$$

Cox-Snell Pseudo R²

Doing the calculations for the next two steps is considerably more difficult. It requires a value for the so-called Cox-Snell pseudo- R^2 (R_{CS}^2), which is a general goodness-of-fit measure for the model we want to fit. For a general summary of the approaches to finding a reasonable R_{CS}^2 , please see Figure 5 in Riley et al. [3].

We might be familiar with the R^2 from ordinary linear regression, where it is a measure between 0-1 of how much the model explains the variation in the data.

We are in the situation where we already have the data and can fit a tentative model. When developing the PBC model we started by fitting a 'maximal' model—a model which was the most complex model we were willing to entertain. We can use that fitted model to calculate

its R_{CS}^2 and then assume that our actual, more realistic model would have some percentage of that R_{CS}^2 , e.g. 80%.

The Cox-Snell pseudo- R^2 has the formula

$$R_{CS}^2 = 1 - \exp\left(2 \frac{\ln L_0 - \ln L_m}{n}\right),$$

where $\ln L_0$ is the maximum log-likelihood of the intercept-only null model and $\ln L_m$ is the maximum log-likelihood of the model. $-2(\ln L_0 - \ln L_m)$ is the log likelihood ratio statistic of the model, and is often reported by statistical software when fitting the model. Knowing this, the ‘maximal’ model in our PBC example has an R_{CS}^2 of

$$R_{CS}^2 = 1 - \exp(-101.95/288) = 0.3$$

of which 80% is 0.24, which we can use as an approximation for what our actual model will give.

In a logistic regression model with event proportion ϕ , $\ln L_0$ is given by

$$\ln L_0 = n(\phi \cdot \ln(\phi) + (1 - \phi) \cdot \ln(1 - \phi))$$

and the maximum value R_{CS}^2 can have for a given ϕ is

$$R_{max}^2 = 1 - \exp\left(2 \frac{\ln L_0}{n}\right) = 1 - \phi^{2\phi}(1 - \phi)^{2(1-\phi)},$$

which has a maximum possible value of 0.75 which occurs for $\phi = 0.5$. This is unlike the ordinary R^2 , which always has a maximum possible value of 1. If we re-scale R_{CS}^2 by dividing it with its maximal possible value based on the event proportion, we get Nagelkerke’s R^2 :

$$R_N^2 = R_{CS}^2 / R_{max}^2,$$

which will have a value between 0 and 1.

Criterion 3

Once we have a usable value for R_{CS}^2 , we can calculate the sample size needed to keep the heuristic shrinkage factor [2] S above e.g. 0.90. The formula is given by

$$n_3 = \frac{p}{(S - 1) \ln\left(1 - \frac{R_{CS}^2}{S}\right)}$$

If we plug in our values ($p = 14$, $R_{CS}^2 = 0.24$) for shrinkage factor $S = 0.90$ we get

$$n_3 = \frac{14}{(0.90 - 1) \ln\left(1 - \frac{0.24}{0.90}\right)} = 455$$

Criterion 4

Lastly, we want the expected optimism of Nagelkerke's R_N^2 to be less than some δ (recommended 0.05). To do this, we first calculate the shrinkage factor S which corresponds with an optimism of δ , which is done according to the formula

$$S = \frac{R_{CS}^2}{R_{CS}^2 + \delta R_{max}^2}$$

Then we just plug this new S into the same formula from Criteria 3, namely

$$n_4 = \frac{p}{(S - 1) \ln \left(1 - \frac{R_{CS}^2}{S} \right)}.$$

In our case we get the new

$$S = \frac{0.24}{0.24 + 0.05 \cdot 0.71} = 0.87,$$

and the needed sample size

$$n_4 = \frac{14}{(0.87 - 1) \ln \left(1 - \frac{0.24}{0.87} \right)} = 338.$$

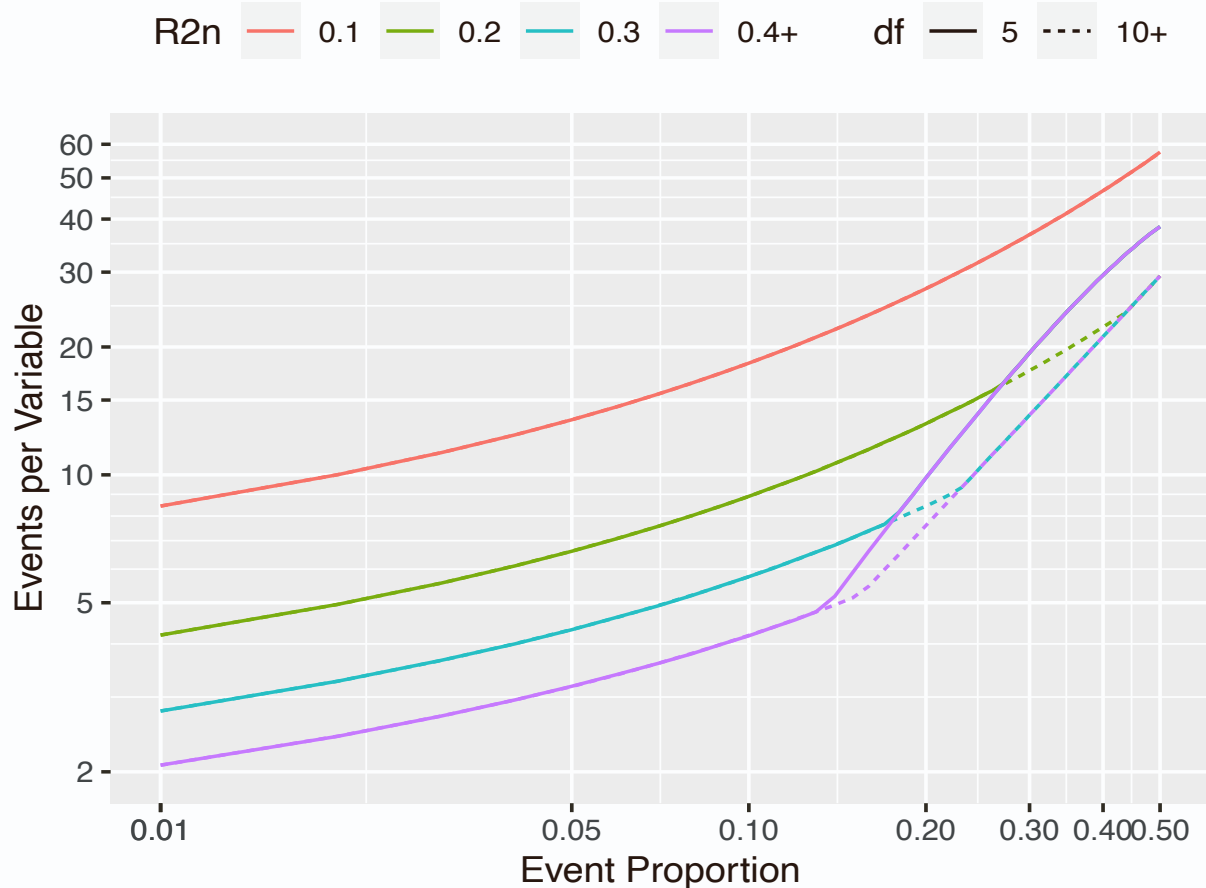
Simplifying the criteria for ease of use

We have 4 good criteria for determining a minimum sample size for developing a prediction model, which is $n = \max(n_1, n_2, n_3, n_4)$. This sample size depends on the (expected) event proportion ϕ , the number of degrees of freedom p we expect to use, and some prior estimate of R_{CS}^2 (or R_N^2).

We will now present a simplified figure for easier and quick reference to the needed sample size. It uses the default recommended values mentioned above for the criteria in terms of event proportion error ($= \pm 0.05$), MAPE ($= 0.05$), shrinkage factor ($= 0.90$), and optimism ($= 0.05$).

If we look at the formulas for n_2, n_3, n_4 they are all proportional to the number of degrees of freedom p (with n_2 proportional to $p^{504/544}$). We can rephrase the sample size criteria into events per variable (EPV, like the old rule of thumb) by multiplying by ϕ and dividing by p . The needed EPV should be approximately the same for all values of p . We found this to be the case for $p \geq 10$. We also found that the EPV was about the same for values of $R_N^2 \geq 0.4$. We then plot the event proportion on the x-axis and the EPV needed on the y-axis, with a line for different levels of R_N^2 and p .

Note that both axes are on $\log_2$ scale.



To use this figure, start by determining the event proportion, either through the data if you already have it, or from subject-matter knowledge of the outcome and the study population. Then determine the expected R^2_N based on how well you expect the model to fit. Then read the EPV needed on the corresponding line. Read from the solid line if you intend to use 10 or more degrees of freedom, and read the dashed line if you expect to use less than 10. From the EPV you can then get the number of events needed by multiplying with the degrees of freedom, and the sample size needed by further dividing by the event proportion.

For reference, the PBC model with 15 degrees of freedom and before shrinkage has $R^2_N = 0.41$, which means we should look at the purple line; and with more than 10 degrees of freedom, we should look at the dashed line. At our event proportion of 0.30 we can read that the needed EPV is 14. If we multiply this by 15 degrees of freedom, we get the minimum number of events needed as 210, which when divided by the event proportion 0.30 gives us the minimum sample size of 700.

Stepwise Simulation Example

The following is a quick simulation example of a logistic regression with a sample size of 2000 and an event proportion of 0.3, where we generate 50 predictors (25 continuous and 25 binary) which are completely independent of the binary outcome. When we apply stepwise selection on these predictors using AIC as the selection criterion, we end up with 7 predictors selected. This model appears to predict the outcome decently well according to the calibration plot. However, if we then generate a new validation data set, all the predictive power is gone, and the calibration curve shows a catastrophic amount of overfitting.

```
require(mvtnorm)
require(ggplot2)

set.seed(123)

N <- 2e3 # sample size
p <- 0.3 # outcome proportion

Y <- rbinom(N, 1, p) # simulate binary outcome

P <- 20 # number of predictors of each type

# Simulate binary predictor values
Xb <- matrix(rbinom(N * P, 1, seq(0.1, 0.9, length = P)),
            nrow = N, ncol = P, byrow = TRUE)

# Variance-covariance matrix for continuous predictors
corr <- diag(1, P, P)
offdiag <- diag(runif(P-1, -0.5, 0.5))
corr <- corr + rbind(cbind(0, offdiag), 0) + rbind(0, cbind(offdiag, 0))
vcov <- corr * sqrt(1:P) %*% t(sqrt(1:P))

# Simulate continuous predictor values
Xc <- rmvnorm(N, sigma = vcov)

# Combine into dataframe
d <- as.data.frame(cbind(Y, Xc, Xb))

# Fit logistic model with all predictors
full_fit <- glm(Y~., d, family = "binomial")

# Perform backwards stepwise selection based on AIC
sw <- step(full_fit, trace = 0)
summary(sw)
```

```
##
## Call:
## glm(formula = Y ~ V2 + V3 + V13 + V19 + V21 + V39 + V40, family =
"binomial",
##      data = d)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.40404      0.16663  -2.425   0.0153 *
## V2           -0.13114      0.05288  -2.480   0.0131 *
## V3           -0.07685      0.03735  -2.058   0.0396 *
## V13          -0.02478      0.01418  -1.748   0.0805 .
## V19          -0.01873      0.01176  -1.592   0.1113
## V21          -0.02200      0.01124  -1.958   0.0502 .
## V39          -0.20670      0.12759  -1.620   0.1052
## V40          -0.34538      0.14045  -2.459   0.0139 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 2429.7  on 1999  degrees of freedom
## Residual deviance: 2404.6  on 1992  degrees of freedom
## AIC: 2420.6
##
## Number of Fisher Scoring iterations: 4

# Get predicted and smoothed observed risks for calibration
pred <- predict(sw, type = "response")
obs <- lowess(pred, d$Y, iter = 0)

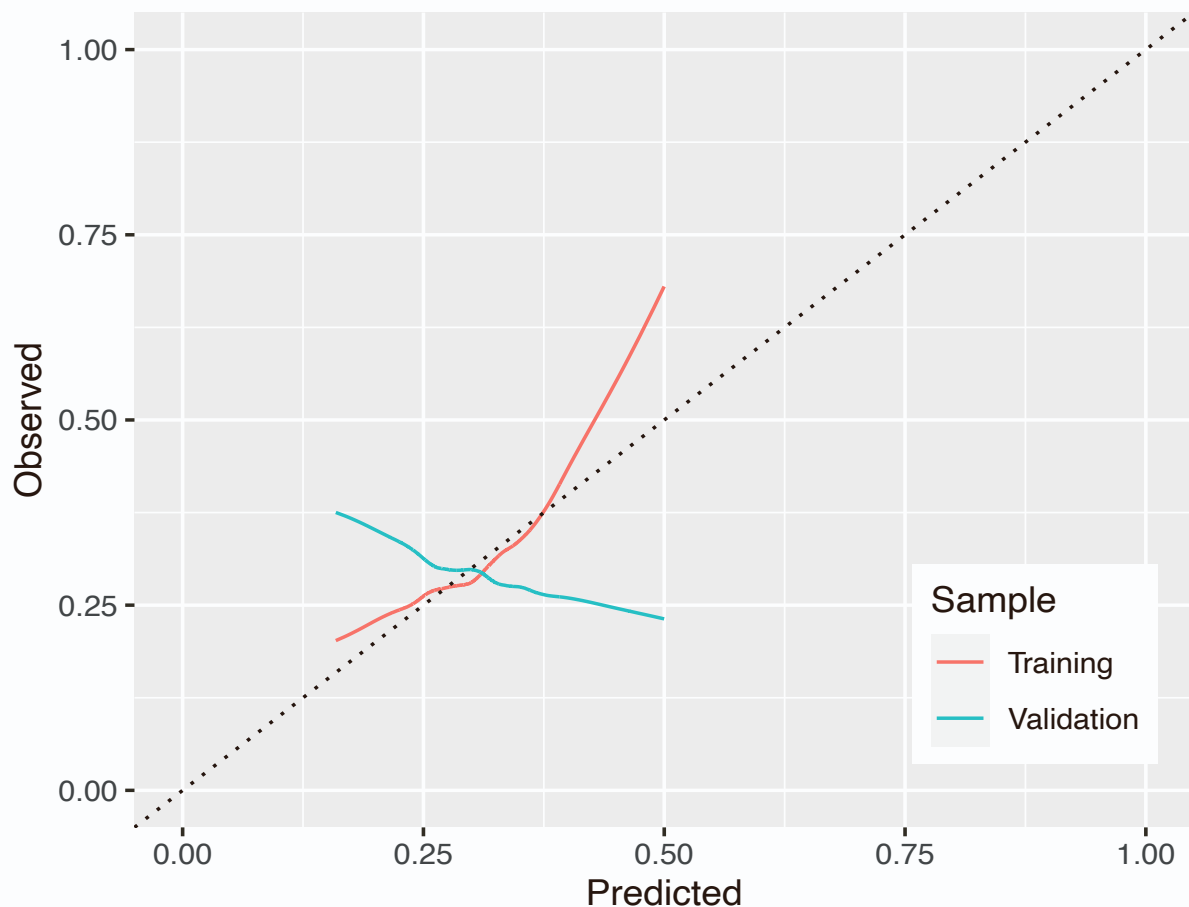
# Generate new sample using the same data generation
Y <- rbinom(N, 1, p)
Xb <- matrix(rbinom(N * P, 1, seq(0.1, 0.9, length = P)),
            nrow = N, ncol = P, byrow = TRUE)
Xc <- rmvnorm(N, sigma = vcov)
d2 <- as.data.frame(cbind(Y, Xc, Xb))

# Get predicted and smoothed observed risks
# for calibration in the new sample
pred2 <- predict(sw, newdata = d2, type = "response")
obs2 <- lowess(pred, d2$Y, iter = 0)

# Plot the calibration curves for the training and validation data
require(ggplot2)
cal_df <- data.frame(Predicted = c(obs$x, obs2$x),
                    Observed = c(obs$y, obs2$y),
                    Sample = rep(c("Training", "Validation"), each = N))

ps3 <- ggplot(cal_df, aes(Predicted, Observed, color = Sample)) +
```

```
geom_line() +
geom_abline(intercept = 0, slope = 1, linetype = 3) +
xlim(0, 1) + ylim(0, 1) +
theme(legend.position = c(0.85, 0.2))
ggsave("figs/fig_s3.pdf", device = cairo_pdf, height = 5, width = 5)
ps3
```



References

- [1] Harrell FE. Regression Modeling Strategies. vol. 13. Cham: Springer International Publishing; 2015. <https://doi.org/10.1007/978-3-319-19425-7>.
- [2] Van Houwelingen JC, Le Cessie S. Predictive value of statistical models. Statistics in Medicine 1990;9:1303–25. <https://doi.org/10.1002/sim.4780091109>.
- [3] Riley RD, Ensor J, Snell KIE, Harrell FE, Martin GP, Reitsma JB, et al. Calculating the sample size required for developing a clinical prediction model. The BMJ 2020;368:1–12. <https://doi.org/10.1136/bmj.m441>.
- [4] Moons KGM, Donders ART, Steyerberg EW, Harrell FE. Penalized maximum likelihood estimation to directly adjust diagnostic and prognostic prediction models for

overoptimism: A clinical example. *Journal of Clinical Epidemiology* 2004;57:1262–70.
<https://doi.org/10.1016/j.jclinepi.2004.01.020>.